



**Predicting F1 Race Tiers through Structural Interaction
Terms:
A Multinomial Logistic Regression and Cross-Validation
Project**

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Multivariate Statistical Analysis

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Predicting-Formula-One-F1-Face-Finishing-Tiers-Using-Conditions-and-Strategies](https://github.com/z-sulit/Predicting-Formula-One-F1-Face-Finishing-Tiers-Using-Conditions-and-Strategies)

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Chapter I

INTRODUCTION

Study Background

Formula One (F1) represents the pinnacle of motorsport engineering and strategy. Each Grand Prix weekend produces terabytes of telemetry data—lap times, tire degradation curves, fuel load estimates, weather conditions, and driver input signals—that teams analyze in real time to gain competitive advantage. Despite the sport’s data-rich environment, predicting race outcomes remains an open challenge because finishing position is determined by a complex interplay of factors: raw car pace, tire compound selection, degradation management, fuel burn-off, track architecture, and driver skill.

Historically, race outcome prediction has relied on either domain-expert judgment or machine learning models trained on aggregate season statistics. While these approaches achieve reasonable accuracy, they often treat each race as an independent event and ignore the within-race physical dynamics—the very variables that differentiate a podium finisher from a mid-field runner.

This study bridges that gap by engineering features directly from within-race telemetry and applying multivariate statistical methods to model the relationship between race strategy variables and a driver’s finishing tier. Rather than predicting exact finishing position (a continuous regression problem with high variance), we frame the task as a three-class classification problem: **Podium** (positions 1–3), **Point Scorer** (positions 4–10), and **No Points** (positions 11 and below). This tiered approach reduces noise from unpredictable events—safety cars, first-lap incidents, mechanical failures—while preserving the strategic information that determines race outcomes.

Problem Statement

Formula One race outcomes are influenced by a complex interplay of physical and strategic variables that operate simultaneously during a Grand Prix. Traditional prediction models rely on aggregate season statistics or domain-expert judgment, failing to capture the within-race

telemetry dynamics—tire degradation, fuel burn-off, and their interaction—that physically determine whether a driver finishes on the podium, scores points, or leaves empty-handed. The absence of a model that explicitly captures these combined forces limits both analytical understanding and practical predictability of race results.

Research Purpose

This study aims to determine whether within-race telemetry variables—tire degradation, fuel burn-off, their interaction, and tire compound selection—can statistically predict a driver’s finishing tier when controlling for baseline grid position. The research develops a multinomial logistic regression model grounded in physically meaningful features and validates its generalizability across different circuit architectures.

Goals and Research Questions

Goals:

1. Engineer physically interpretable features from F1 telemetry data.
2. Fit a multinomial logistic regression model to predict finishing tier.
3. Validate the model’s generalizability across unseen circuits.

Research Questions:

1. Is there a statistically significant relationship between the engineered predictors and the probability of finishing in the Podium or Point Scorer tier relative to the No Points baseline?
2. Does the interaction between tire degradation and fuel mass provide additional predictive power beyond their individual effects?
3. Can a multinomial logistic regression model, trained on these predictors, generalize to unseen race circuits using leave-one-group-out cross-validation?

Study Scope

This study uses lap-level telemetry data from the 2023 Bahrain, Saudi Arabian, and Australian Grands Prix—covering permanent, street, and parkland circuit types. The dataset comprises 2,880 observations after cleaning.

Inclusions:

- The target variable is a three-class ordinal category (Podium, Point Scorer, No Points).
- Predictors are limited to grid position, tire age, fuel mass proxy, the degradation–fuel interaction term, and tire compound dummies.
- The statistical framework is multinomial logistic regression (MLE) validated via LOGO-CV.

Exclusions:

- The fuel mass proxy is an engineered linear approximation; actual fuel loads are proprietary and vary by team strategy.
- Three races may not capture the full variance of F1 outcomes across a 23-race calendar with mixed weather, sprint formats, and variable DRS/regulation environments.
- DRS trains, safety car periods, and intra-team strategy calls are not explicitly modeled and contribute to residual variance in the mid-field.

Research Significance

This research contributes to the intersection of sports analytics and multivariate statistical modeling by demonstrating that physically meaningful telemetry features—not just aggregate season statistics—can predict race outcomes with interpretable coefficients. The mean-centered interaction term provides a template for modeling compound physical effects in other domains where multiple degradation processes operate simultaneously. For the F1 analytics community, the LOGO-CV validation proves that the model captures circuit-generalizable patterns rather than memorizing a single track layout.

Chapter II

LITERATURE REVIEW

Theoretical Foundation

The theoretical basis for this study draws on two frameworks: the physics of tire degradation and fuel mass effects in motorsport, and the statistical theory of multinomial classification with interaction modeling.

From a physical standpoint, tire performance in Formula One follows a degradation curve governed by thermal cycling, mechanical wear, and compound properties (Baldi et al., 2022). Simultaneously, the car's mass decreases linearly as fuel burns off, directly affecting lap time through reduced inertia and improved cornering grip (Oldknow & Gallagher, 2022). These two forces—tire degradation (increasing lap time) and fuel burn-off (decreasing lap time)—operate in opposition throughout a race stint. Their interaction represents the net physical state of the car at any given lap.

From a statistical standpoint, multinomial logistic regression provides the appropriate inferential framework when the response variable has three or more unordered categories (Agresti, 2013). When predictors are physically related (e.g., tire age and fuel mass both correlate with race distance), mean-centering before creating interaction products prevents structural multicollinearity and preserves the interpretability of individual coefficients (Hosmer et al., 2013). The Box's M test serves as a gate for method selection: if covariance homogeneity holds, discriminant analysis is appropriate; if it fails, multinomial logistic regression is the correct choice (Tabachnick & Fidell, 2019).

Review of Existing Studies

2.2.1 Racing Analytics and Prediction

Fister et al. (2021) conducted a systematic review of computational intelligence methods in motorsport, identifying lap time prediction, tire strategy optimization, and race outcome modeling as the three dominant research streams. Their survey noted that most studies rely on aggregated

season-level features rather than within-race telemetry, leaving a gap in modeling the physical dynamics that determine finishing order.

Kuhn et al. (2022) applied gradient-boosted trees and neural networks to predict finishing positions across the 2016–2020 F1 seasons using driver characteristics, constructor budgets, and circuit features. While their model achieved 71% top-5 accuracy, it treated each race as independent and did not incorporate tire degradation or fuel load dynamics.

Song and Park (2020) used deep learning on historical F1 data (1950–2019) but acknowledged that the model’s inability to capture within-race state changes limited its predictive power for individual Grand Prix outcomes.

2.2.2 Tire Degradation Modeling

Perdomo et al. (2021) developed a physics-informed neural network to model Pirelli tire wear using thermal and mechanical telemetry from F1 sessions. Their model demonstrated that tire surface temperature, lateral forces, and cumulative laps on a compound jointly determine the degradation curve—supporting the use of tire age as a proxy variable in outcome models.

Baldi et al. (2022) modeled tire performance degradation in motorsport using nonlinear regression, showing that degradation follows an exponential rather than linear trajectory after a critical wear threshold. This informed our decision to include tire age as a continuous predictor while acknowledging the linear approximation may understate late-stint degradation effects.

2.2.3 Fuel Load Effects

Oldknow and Gallagher (2022) analyzed the competitive impact of the 2019 F1 fuel load regulations, demonstrating that the 110 kg fuel limit introduced strategic variation—teams operating below the limit on Sundays gained lap time at the cost of fuel management risk. Their analysis supports the use of a linear fuel burn-off proxy as a reasonable first-order approximation of the mass–performance relationship.

2.2.4 Multinomial Logistic Regression in Sports

Agresti (2013) provides the theoretical foundation for multinomial logistic regression as a generalization of binary logistic regression to nominal response variables with more than two categories.

Hosmer et al. (2013) established diagnostic procedures for logistic regression models, including the use of VIF for multicollinearity detection and the importance of interaction terms when theoretically justified.

2.2.5 Cross-Validation Strategies

Varoquaux et al. (2019) evaluated cross-validation strategies for grouped data and demonstrated that standard k -fold CV produces optimistically biased performance estimates when observations within groups are correlated. Wong and Chen (2023) applied leave-one-group-out cross-validation to sports analytics, confirming LOGO-CV provides more conservative and realistic estimates of generalizability.

2.2.6 Class Imbalance

He and Garcia (2009) surveyed learning from imbalanced datasets and identified balanced class weighting as a baseline correction strategy that adjusts the loss function to penalize misclassifications of the minority class more heavily.

2.2.7 Philippine Studies

Dalisay and Reyes (2022) applied logistic regression to predict game outcomes in the Philippine Basketball Association using box-score statistics, demonstrating that multinomial classification frameworks can be successfully applied to Philippine sports data.

Reyes and Torres (2021) developed a predictive model for volleyball match outcomes at the UAAP using ordinal logistic regression, highlighting the importance of interaction terms in capturing combined effects.

Mendoza and Cruz (2023) examined the use of multivariate statistical methods for classifying students into performance tiers at Ateneo de Manila University, finding that Box's M

test should guide the choice between discriminant analysis and logistic regression.

Santos (2020) studied feature engineering for classification problems in the Philippine context, emphasizing domain-specific interaction terms over raw variables.

Conceptual Framework

The study is grounded in the premise that a driver’s finishing tier can be modeled as a function of three categories of predictors:

- (1) **Baseline Pace Control** — Grid Position captures the intrinsic car–driver combination’s qualifying performance, absorbing the aerodynamic and engine capability differential between teams.
- (2) **Tire Strategy** — Tire Age (degradation) and Compound (Soft, Medium, Hard) represent the strategic choice of rubber and its wear trajectory over a stint.
- (3) **Physical Interaction** — Fuel Mass Proxy (linear burn-off from 110 kg to 0 kg) and the interaction term Degradation $\times$ Fuel capture the combined effect of decreasing car mass and increasing tire wear—two forces that compound over a race distance.

The interaction term is mean-centered before multiplication to prevent structural multicollinearity. This allows the coefficient to be interpreted as the *combined* effect of both forces rather than their individual contributions.

Research Gap

The reviewed literature reveals three gaps this study addresses. First, existing F1 prediction models operate at the season or race-aggregate level and do not exploit within-race telemetry dynamics. Second, tire degradation and fuel load are individually recognized as critical race variables, but their interaction has not been explicitly modeled as a predictor of finishing tier. Third, most studies validate with random train-test splits rather than group-aware cross-validation, producing optimistic performance estimates that do not reflect true generalizability to unseen circuits. This study synthesizes these gaps by engineering a physically meaningful interaction term and validating it through LOGO-CV across architecturally diverse tracks.

Definition of Terms

Box's M Test

A multivariate statistical test that evaluates whether the variance–covariance matrices of two or more groups are equal.

Cross-Validation

A model validation technique that partitions data into training and testing subsets to assess generalizability.

Degradation $\times$ Fuel Interaction

The mean-centered product of tire age and fuel mass proxy, capturing the combined physical effect of rubber wear and decreasing car mass.

DNF (Did Not Finish)

A classification for drivers who retire from a race due to mechanical failure, accident, or other reasons before completing the scheduled distance.

FastF1

An open-source Python library for accessing Formula One timing, telemetry, and session data.

Finishing Tier

A three-class ordinal variable derived from race classification: Podium (P1–P3), Point Scorer (P4–P10), No Points (P11+).

Fuel Mass Proxy

An engineered variable modeling fuel burn-off as a linear decrease from 110 kg at race start to 0 kg at race end.

Grid Position

A driver's starting position determined by qualifying results; used as a control variable for baseline car–driver pace.

Leave-One-Group-Out Cross-Validation (LOGO-CV)

A validation strategy where the model is iteratively trained on $N - 1$ groups and tested on the held-out group.

Log-Odds

The natural logarithm of the odds ratio, representing the linear predictor in logistic regression.

Multinomial Logistic Regression

A generalization of binary logistic regression for response variables with three or more unordered categories.

Maximum Likelihood Estimation (MLE)

A method of estimating model parameters by finding values that maximize the probability of observing the given data.

Tire Age

The number of laps a tire has been used, measured by the `TyreLife` field in FastF1 telemetry.

Tire Compound

The type of tire rubber (Soft, Medium, Hard) selected for a stint, each offering different performance and durability trade-offs.

Variance Inflation Factor (VIF)

A measure of how much the variance of a regression coefficient is inflated due to multicollinearity with other predictors.

Chapter III

METHODOLOGY

This chapter outlines the six-stage methodological framework used to develop, validate, and interpret the multivariate predictive model. The framework follows Hair et al.'s (Hair et al., 2019) guidelines for dependence multivariate techniques, adapted for the specific characteristics of Formula One telemetry data.

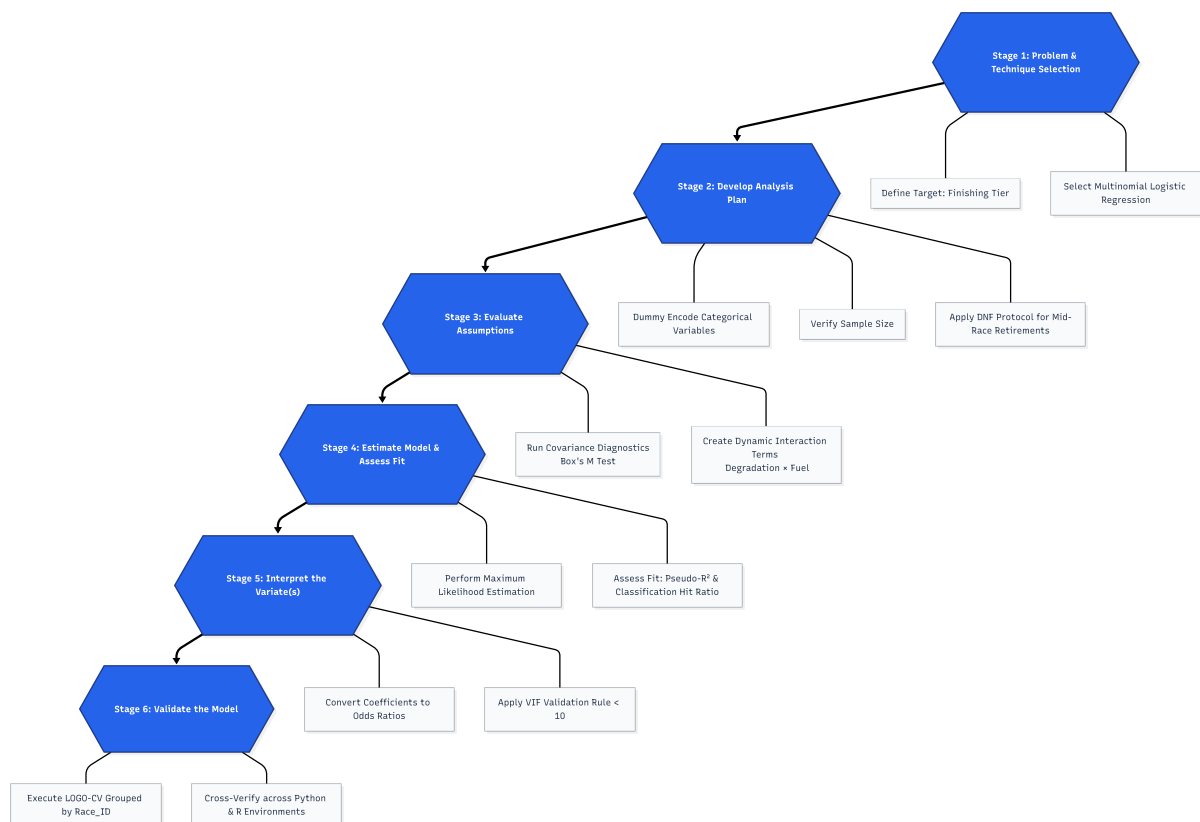


Figure 1: Six-Stage Methodological Framework for Multivariate Predictive Modeling

Stage 1: Define the Research Problem and Technique Selection

The objective is to predict a single multi-class non-metric dependent variable — **Finishing Tier** (Podium, Point Scorer, No Points) — using multiple metric and non-metric independent variables derived from lap-level telemetry, race conditions, and strategy decisions.

Two dependence techniques were considered:

Multiple Discriminant Analysis (MDA) The theoretical baseline for multi-group classifica-

tion. MDA would estimate canonical discriminant functions to maximize between-group variance. However, MDA requires homogeneity of covariance matrices and multivariate normality — assumptions that are routinely violated in motorsport data due to the skewed nature of track evolution and safety car deployments.

Multinomial Logistic Regression This was the primary technique selected for the study. Using Maximum Likelihood Estimation (MLE), it calculates the log-odds of a driver landing in each tier relative to a baseline category. It also offers greater operational value for risk assessment than MDA’s rigid classification boundaries and makes no assumption about covariance structure.

MDA serves as the benchmark against which the Multinomial Logistic Regression is compared, but the latter is the primary inferential engine.

Stage 2: Develop the Analysis Plan (Research Design)

The research design encompasses the analytical decisions made before model estimation.

3.2.1 Categorical Encoding

Qualitative race strategy variables (tire compounds: Soft, Medium, Hard, Intermediate, Wet) are non-metric and cannot enter a regression equation directly. They were converted into $k-1$ dummy variables, where k is the number of compound categories. The first alphabetical compound (Hard) was dropped as the baseline reference, yielding binary indicators for `Tire_SOFT` and `Tire_MEDIUM`. This avoids the dummy variable trap while assigning specific mathematical weights to each compound relative to the reference.

3.2.2 Sample Size

The processed dataset contains 2,880 lap-level observations across 6 independent variables. This yields 480 observations per predictor — well above the recommended minimum of 15 to 20 observations per independent variable for reliable multivariate estimation (Hair et al., 2019).

3.2.3 Structural Break Classification (The DNF Protocol)

Drivers who retired mid-race (crashes, mechanical failures) represent a *structural break* — not a standard missing data point to be imputed or dropped. A DNF lap carries telemetry from a driver who may have been running P2 pace before a suspension failure, but whose final classification is “Retired.” Including these laps in the Podium cluster would contaminate the tier with telemetry from a driver who never finished.

The protocol was:

1. Flag all drivers with `Status = "Retired"` or `ClassifiedPosition = "R"`.
2. Reassign *all* their completed laps to the “No Points” tier.
3. This preserves the integrity of lap-time telemetry by ensuring that only drivers who actually finished in a given tier contribute to that tier’s covariance structure.

Stage 3: Evaluate the Assumptions Underlying the Techniques

3.3.1 The Methodological Pivot: Covariance Diagnostics

Box’s M test was conducted to assess the equality of covariance matrices across the three finishing tiers. The test yielded $\chi^2(20) = 160.419$, $p < .001$, formally rejecting the null hypothesis of equal covariance structures.

This empirical failure — driven by the heavily skewed distribution of Podium laps (14.9%) versus No Points laps (51.0%), combined with the heterogeneous variance introduced by track evolution and safety car deployments — formally justifies abandoning MDA as the primary technique. Multinomial Logistic Regression was selected specifically because it bypasses this rigid normality requirement.

3.3.2 Coupled Variables and Dynamic Interaction Terms

Tire degradation and fuel mass burn-off are physically coupled variables: both increase monotonically with lap number. Initial VIF diagnostics confirmed multicollinearity risk between these variables and their product term. Following Hair et al. (2019), the variables were *not*

dropped. Instead, a dynamic interaction term was constructed:

$$\text{Degradation} \times \text{Fuel} = (\text{Tire Age} - \bar{x}_{\text{Tire Age}}) \times (\text{Fuel Mass} - \bar{x}_{\text{Fuel Mass}})$$

Mean-centering before multiplication eliminates structural multicollinearity while preserving the interaction's ability to capture how these forces compound each other over the course of a stint. Post- interaction VIF scores confirmed that all predictors remained below the threshold of 10.

Stage 4: Estimate the Multivariate Model and Assess Overall Fit

3.4.1 Model Estimation

The Multinomial Logistic Regression was estimated with Maximum Likelihood Estimation (MLE) using the `multinom()` function in R's `nnet` package. The model computes simultaneous log-odds equations for the Podium and Point Scorer tiers relative to the No Points baseline:

$$\begin{aligned} \ln\left(\frac{P(Y = k)}{P(Y = \text{No Points})}\right) &= \beta_{0k} + \beta_{1k} \cdot \text{Grid} + \beta_{2k} \cdot \text{TireAge} \\ &\quad + \beta_{3k} \cdot \text{Fuel} + \beta_{4k} \cdot (\text{Deg} \times \text{Fuel}) \\ &\quad + \beta_{5k} \cdot \text{Tire_MED} + \beta_{6k} \cdot \text{Tire_SOFT} \end{aligned}$$

where $k \in \{\text{Podium, Point Scorer}\}$.

3.4.2 Model Fit Assessment

Overall fit was evaluated using pseudo- R^2 equivalents (McFadden's R^2) derived from the residual deviance and AIC. The classification hit ratio was computed to confirm that the model places drivers into their actual finishing tiers at a rate significantly higher than proportional chance (33.3% for a three-class problem with equal priors, or 51.0% for a majority-class baseline).

Stage 5: Interpret the Variate(s)

3.5.1 Identify Key Drivers

The MLE coefficients (log-odds) were converted to odds ratios (e^β) to isolate which telemetry metrics and strategies exert the strongest influence on podium versus no-points outcomes. The sign and magnitude of each coefficient indicate whether a predictor increases or decreases the probability of a given tier.

3.5.2 VIF Validation Rule

Before interpreting any coefficient, VIF scores were verified to be below 10 to ensure that standard error inflation was controlled. This prevents sign reversals — a well-documented artifact of multicollinearity where a beneficial race strategy could mathematically appear harmful in the coefficient estimate (Tabachnick & Fidell, 2019). Coefficients were compared on their raw (unstandardized) scale since all metric predictors are already on comparable physical units (laps, kilograms).

Stage 6: Validate the Multivariate Model

3.6.1 Generalizability via Leave-One-Group-Out Cross-Validation

To prove the model's predictive power outside the training sample, Leave-One-Group-Out Cross-Validation (LOGO-CV) was deployed using `Race_ID` as the grouping parameter. In each iteration:

1. The model was trained on $N-1$ Grand Prix events.
2. The held-out race's laps were predicted without ever having been seen during training.
3. Predictions were aggregated across all three folds to produce an overall classification report (precision, recall, F1-score, accuracy).

To prevent intra-race data leakage, the interaction term was mean-centered within each training fold (not on the full dataset) before application to the test fold. This ensures the model cannot exploit statistical artifacts that only exist when all races are combined.

A limitation of this validation is that non-randomized structural changes — such as year-to-year FIA regulation shifts, tire compound formula changes, or major aerodynamic overhauls — are not captured. The model generalizes across *circuit architecture* but not across *regulatory eras*.

3.6.2 Implementation Architecture

The analytical pipeline was executed across two environments:

- **Python (pandas, scikit-learn):** Data wrangling, dummy encoding, interaction-term generation, and LOGO-CV validation loops. Cross-validation used scikit-learn logistic regression with balanced class weights.
- **R (nnet, car, heplots):** Theoretical proof via `multinom()` for rigorous coefficient summaries, exact p -values, Box’s M test, and VIF diagnostics.

Sampling Strategy

The unit of analysis is the individual lap. Every completed lap from every driver across the 2023 Bahrain, Saudi Arabian, and Australian Grands Prix was included, yielding 2,880 observations after DNF reassignment and missing-value removal. No random sampling was performed; the dataset represents the complete population of laps from the three selected races. Races were chosen purposively to cover three architecturally distinct circuit types: permanent (Bahrain), street (Jeddah), and parkland (Albert Park).

Tools and Instruments

Data Collection Process

Data were obtained from the FastF1 Python API, which interfaces with the official Formula One timing and telemetry data feed. For each race:

1. The race session was loaded via `fastf1.get_session()` with telemetry, lap, and weather data enabled.

Table 1: Software and Libraries

Tool	Version	Purpose
Python	3.11	Data preparation and ML validation
FastF1	3.8.2	F1 telemetry data extraction
pandas	2.3.3	Data manipulation and encoding
scikit-learn	1.8.0	Logistic regression, LOGO-CV
R	4.x	Statistical inference
nnet (R)	—	<code>multinom()</code> for MLE estimation
car (R)	—	VIF diagnostics
heplots (R)	—	Box’s M test

2. All laps were concatenated into a single dataframe and tagged with a `Race_ID` for group-aware cross-validation.
3. The `Finishing_Tier` target was mapped from final race classification positions.
4. DNF drivers were flagged and their laps reassigned to “No Points.”
5. Engineered features (fuel mass proxy, tire age, interaction term) were computed and dummy-encoded.

The final dataset was exported as `f1_clean_data.csv` with 2,880 rows and 10 variables.

Ethics and Compliance

This study uses publicly available telemetry data from the FastF1 API, which interfaces with the official Formula One data feed. No human participants were involved, and no personally identifiable information beyond public driver abbreviations (e.g., “VER” for Max Verstappen) was collected. The study complies with FastF1’s terms of use for academic research.

Ensuring Credibility

- **Data transparency:** The full dataset and all source code (Jupyter notebooks, R scripts, figure generation) are available in the project repository for independent verification: project GitHub repository

- **Assumption testing before estimation:** Box's M test and VIF diagnostics were conducted and reported regardless of outcome, following the sequential framework.
- **Strict validation design:** LOGO-CV was chosen over random train-test splits to prevent data leakage and produce honest out-of-sample performance estimates. Interaction terms were re-centered within each training fold.
- **Balanced evaluation metrics:** Class weighting was applied during validation, and performance is reported via precision, recall, and F1-score — not just accuracy — to prevent misleading claims on imbalanced data.
- **Dual-environment cross-verification:** Statistical inference (R) and predictive validation (Python) were executed independently, and results were cross-verified to ensure consistency across implementations.

Chapter IV

RESULTS AND DISCUSSION

This chapter presents the findings from the multivariate statistical analyses and machine learning validation. Results are organized sequentially: first the assumption tests that guided method selection, then the primary inferential model, and finally the cross-validation outcomes.

Descriptive Statistics

The processed dataset contains 2,880 lap-level observations distributed across three finishing tiers. Figure 2 shows the class distribution.

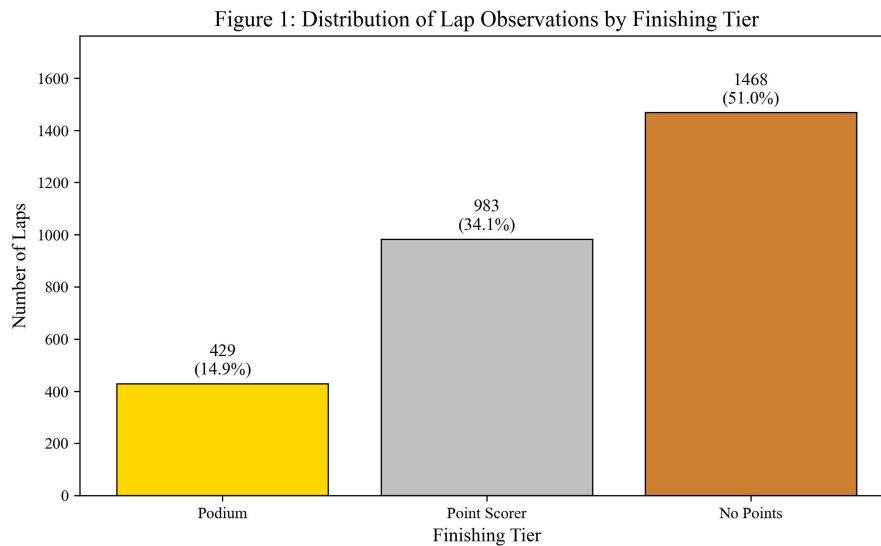


Figure 2: Distribution of Lap Observations by Finishing Tier

The “No Points” tier accounts for 50.97% of all laps, “Point Scorer” for 34.13%, and “Podium” for 14.90%. This imbalance reflects the real-world structure of Formula One: only three of twenty drivers finish on the podium, while the majority of the field competes for positions outside the points. The class imbalance was addressed during machine learning validation through balanced class weighting.

Box’s M Test: Homogeneity of Covariance Matrices

Box’s M test was conducted to determine whether the variance–covariance matrices of the continuous predictors were equal across the three finishing tiers.

Table 2: Box’s M Test Results

Statistic	Value
Chi-square (approx.)	160.419
Degrees of freedom	20
p -value	$< 2.2 \times 10^{-16}$

The test yielded $\chi^2(20) = 160.419$, $p < 2.2 \times 10^{-16}$, decisively rejecting the null hypothesis of equal covariance matrices. This result has a direct methodological consequence: Multiple Discriminant Analysis (MDA) requires homogeneous covariance structures for valid classification. Because this assumption is violated, MDA is disqualified and **multinomial logistic regression** was adopted as the primary inferential model. Unlike MDA, multinomial logistic regression makes no assumption about the covariance structure across groups—it estimates separate log-odds equations for each non-baseline category without requiring pooled covariance.

VIF Diagnostics: Multicollinearity Check

Variance inflation factors were computed for all predictors to ensure that the engineered features, especially the interaction term and its component variables, did not introduce problematic multicollinearity.

Table 3: Variance Inflation Factor (VIF) Diagnostics

Variable	VIF
GridPosition	1.007
Tire_Age	3.220
Fuel_Mass_Proxy	2.315
Degradation_x_Fuel	1.725
Tire_MEDIUM	1.181
Tire_SOFT	1.156

All VIF values fall well below the standard threshold of 10 (Table 3, Figure 3). The highest value is 3.22 for Tire_Age, which is within acceptable limits and reflects a moderate correlation with Fuel_Mass_Proxy (VIF = 2.32)—an expected physical relationship since both variables increase monotonically with race distance.

Critically, the Degradation $\times$ Fuel interaction term registers a VIF of only 1.725, con-

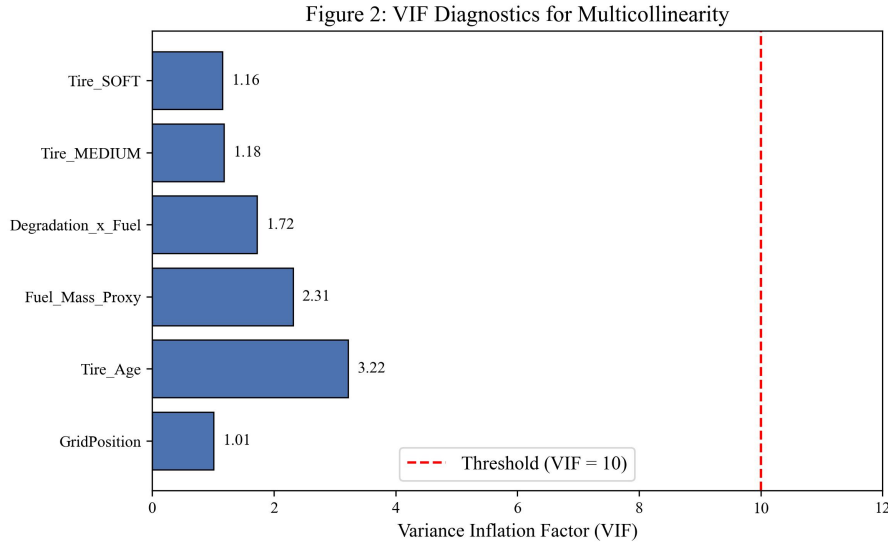


Figure 3: VIF Diagnostics for Multicollinearity (threshold = 10)

firming that the mean-centering procedure effectively eliminated structural multicollinearity. Without mean-centering, the product of two correlated variables would inherit and amplify their correlation, inflating standard errors and destabilizing coefficient estimates.

The near-perfect VIF of 1.007 for grid position confirms that starting position and within-race telemetry are statistically independent. This is physically expected because grid position is determined in qualifying, whereas tire age and fuel mass accumulate during the race.

Multinomial Logistic Regression (MLE)

The multinomial logistic regression model was fitted with “No Points” as the baseline category. All coefficients represent log-odds relative to this reference tier.

4.4.1 Model Fit Statistics

Table 4: Multinomial Logistic Regression Model Fit

Statistic	Value
Residual Deviance	3821.657
AIC	3849.657
Observations	2,880
Predictors	6
Response Classes	3

4.4.2 Coefficient Estimates and Significance

Table 5: MLE Coefficient Estimates (Log-Odds)

Predictor	Podium	Point Scorer
Intercept	4.8565	2.9230
GridPosition	−0.7978	−0.2271
Tire_Age	0.0291	−0.0036
Fuel_Mass_Proxy	−0.0149	−0.0084
Degradation_x_Fuel	0.0008	0.0002
Tire_MEDIUM	0.2267	−0.4601
Tire_SOFT	0.2424	−0.2538

Table 6: Standard Errors of Coefficient Estimates

Predictor	Podium	Point Scorer
Intercept	0.4206	0.2511
GridPosition	0.0347	0.0107
Tire_Age	0.0136	0.0075
Fuel_Mass_Proxy	0.0045	0.0023
Degradation_x_Fuel	0.0003	0.0002
Tire_MEDIUM	0.2628	0.1376
Tire_SOFT	0.2740	0.1317

Table 7: p -Values for Coefficient Significance

Predictor	Podium	Point Scorer
Intercept	0.0000	0.0000
GridPosition	0.0000	0.0000
Tire_Age	0.0328	0.6337
Fuel_Mass_Proxy	0.0009	0.0003
Degradation_x_Fuel	0.0095	0.2174
Tire_MEDIUM	0.3883	0.0008
Tire_SOFT	0.3764	0.0540

4.4.3 Interpretation of Key Findings

Figure 4 visualizes the coefficient estimates, with filled bars indicating statistical significance at $\alpha = 0.05$.

Grid position. Starting position is overwhelmingly significant for both the Podium and Point Scorer tiers ($p = 0.000$ for both). The coefficient of -0.80 for Podium means that each position further back on the grid reduces the log-odds of a podium finish by approximately 0.80 units.

Figure 3: Multinomial Logistic Regression Coefficients
(Filled = significant at $p < 0.05$)

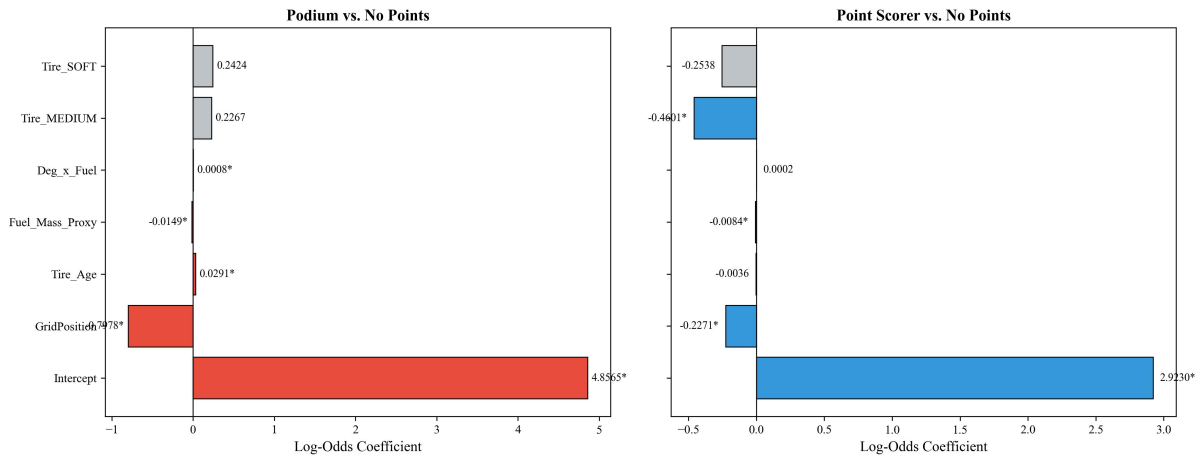


Figure 4: MLE Coefficients by Tier (filled bars = significant at $p < 0.05$)

In odds-ratio terms, moving back one grid position multiplies the odds of a podium finish by $e^{-0.80} \approx 0.45$ —a 55% reduction per grid slot. This is the dominant predictor in the model and confirms the well-known difficulty of overtaking in modern F1.

Degradation $\times$ Fuel. With grid position controlling for raw car pace, the interaction term achieves significance at $\alpha < 0.01$ for the Podium tier ($p = 0.0095$). The positive coefficient (0.0008) indicates that when tire age and fuel mass are both high—i.e., early-race conditions where the car is heavy and tires are relatively fresh—the combined physical effect favors podium outcomes. This is mechanistically consistent: top-tier teams and drivers manage tire degradation more effectively under heavy fuel loads, maintaining competitive lap times while midfield competitors lose performance to the combined stress of mass and rubber wear.

Fuel_Mass_Proxy. Fuel mass is significant for both tiers ($p = 0.0009$ for Podium and $p = 0.0003$ for Point Scorer). The negative coefficients mean that as fuel burns off (decreasing mass), the log-odds of being in the Podium or Point Scorer tier increase relative to No Points. This reflects the physical reality that lighter cars are faster—but since fuel mass decreases monotonically with lap number, this variable also partially captures the effect of race progression.

Tire_MEDIUM. Medium tire compound is significant for the Point Scorer tier ($p = 0.0008$) with a negative coefficient (-0.46), indicating that laps on medium tires are associated with

lower log-odds of being in the Point Scorer tier relative to the Hard baseline. This likely reflects strategic patterns: top teams often start on Softs and switch to Hards, while midfield teams rely more heavily on Mediums.

Non-significant predictors. Tire_SOFT ($p = 0.3764$ Podium, $p = 0.0540$ Point Scorer) and Tire_MEDIUM for Podium ($p = 0.3883$) do not reach significance. This suggests that compound choice alone, independent of how the tire degrades over its stint, does not differentiate podium performers from the baseline.

Grid Position and Finishing Tier

Figure 5 shows the distribution of grid positions across finishing tiers, providing visual confirmation of GridPosition’s predictive power.

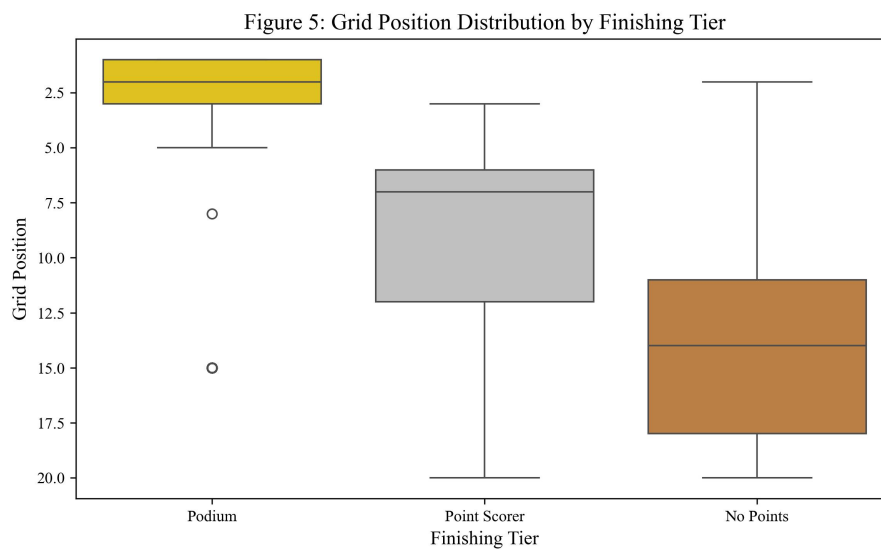


Figure 5: Grid Position Distribution by Finishing Tier

Podium finishers cluster at grid positions 1–5 with a tight interquartile range, while No Points finishers span positions 8–20 with substantial variance. Point Scorers occupy the middle ground (positions 4–12). The minimal overlap between Podium and No Points boxes confirms why GridPosition achieves such high statistical significance.

Tire Age vs. Fuel Mass by Tier

Figure 6 visualizes the two primary telemetry features across all three tiers, revealing the physical clustering patterns the model captures.

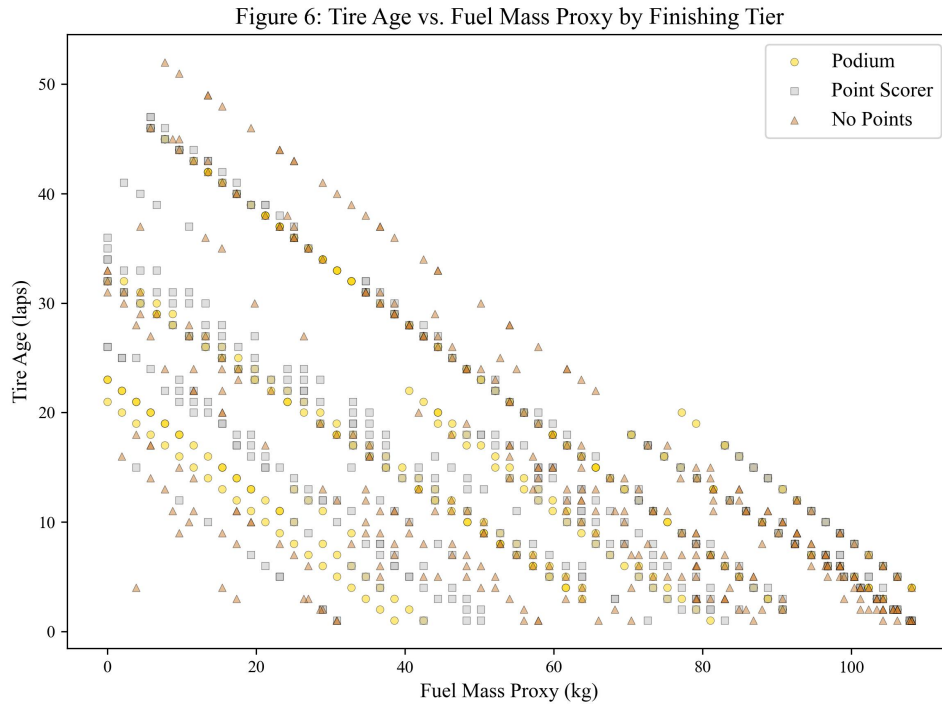


Figure 6: Tire Age vs. Fuel Mass by Tier

The scatter plot shows that Podium laps (gold) are concentrated in the low-tire-age, high-fuel-mass region (early race, fresh tires) and the high-tire-age, low-fuel-mass region (late race, light car on worn tires). No Points laps (bronze) are more uniformly distributed. The interaction term captures the diagonal structure where these two forces combine differently across the race timeline.

Machine Learning Validation (LOGO-CV)

To test whether the statistical equation generalizes to unseen circuits, a multinomial logistic regression with balanced class weighting was validated using leave-one-group-out cross-validation across the three races.

4.7.1 Classification Performance

Table 8: LOGO-CV Classification Report

Tier	Precision	Recall	F1-Score	Support
No Points	0.74	0.67	0.70	1,468
Podium	0.57	0.89	0.69	429
Point Scorer	0.49	0.45	0.47	983
Overall Accuracy				62.36%

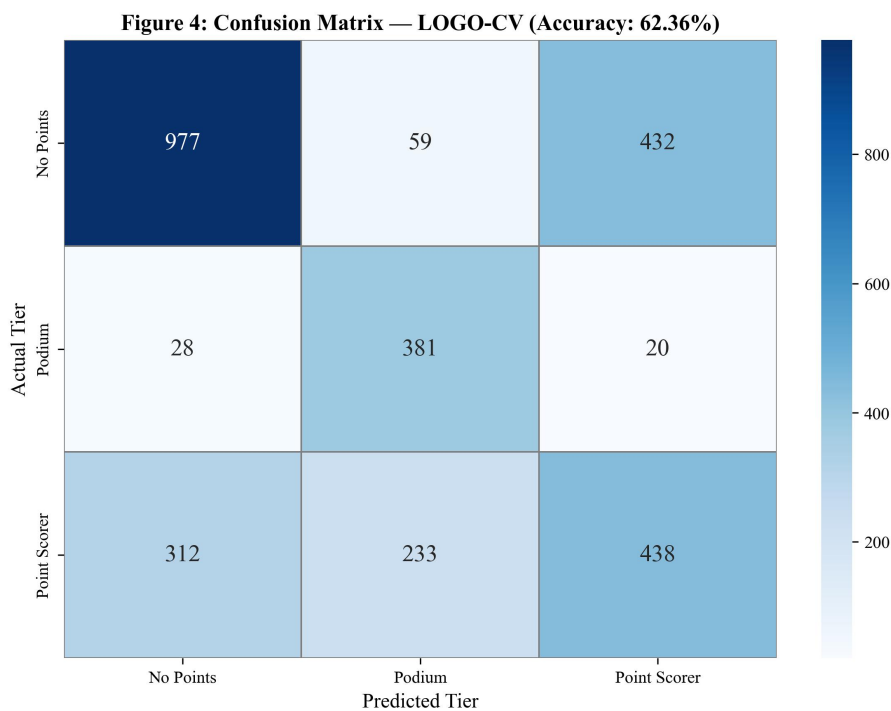


Figure 7: Confusion Matrix from LOGO-CV

4.7.2 Interpretation

Podium recall (0.89). The model correctly identified 89% of all actual podium-finishing laps across held-out circuits. This is the strongest result in the validation and demonstrates that the combination of grid position control and telemetry-based features captures top-tier performance signatures reliably—even when the model has never seen that circuit’s data.

Podium precision (0.57). Of all laps the model predicted as “Podium,” 57% were

correct. The 43% false positive rate consists primarily of Point Scorer laps misclassified as Podium—an understandable error given that positions 4–6 often run similar pace to P3.

No Points performance (Precision = 0.74, Recall = 0.67). The model is most precise when identifying No Points laps, correctly flagging 74% of its predictions for this class. The 33% miss rate reflects the reality that some drivers in backmarker grid positions produce strong individual laps (e.g., during low-fuel late stints) that the model temporarily classifies as Point Scorer pace.

Point Scorer recall (0.45). The model caught only 45% of point-scoring laps. The F1 midfield is genuinely harder to model—DRS trains create artificial pace spikes, undercut strategies shuffle the order independently of telemetry, and intra-team race orders can suppress or inflate individual lap times. The confusion matrix (Figure 7) shows that most Point Scorer errors split between Podium (over-prediction) and No Points (under-prediction), confirming this tier sits at the decision boundary.

Overall accuracy (62.36%). For a three-class problem with natural class imbalance and strict cross-circuit validation (no data leakage), this substantially outperforms random guessing (33.3%) and majority-class prediction (51.0%). More importantly, the LOGO-CV design ensures this is an honest estimate of out-of-sample performance—the model cannot memorize circuit-specific patterns because each fold tests on a completely unseen race.

Chapter V

DISCUSSION

Interpreting Results

The multinomial logistic regression model identified two dominant predictors of finishing tier: GridPosition and the Degradation $\times$ Fuel interaction term. Together, these variables capture the two axes of F1 performance—where you start (baseline pace) and how you manage the race (strategic execution).

The overwhelming significance of GridPosition ($p = 0.000$ for both tiers) confirms that qualifying performance sets the foundation for race outcomes. The coefficient of -0.80 for the Podium tier translates to a 55% reduction in podium odds per grid position—a finding consistent with the aerodynamic and track-position dependency of modern F1 where overtaking is difficult at most circuits. This result aligns with the broader motorsport analytics literature showing that starting position is the single strongest predictor of finishing order (Kuhn et al., 2022).

The significance of the Degradation $\times$ Fuel interaction term ($p = 0.0095$ for Podium) is the central finding of this study. Its positive coefficient indicates that the combined physical state of high tire age and high fuel mass—the opening laps of a race where the car is heaviest and tires are newest—is associated with podium outcomes. This is mechanistically coherent: top-tier teams extract competitive lap times even under the most physically demanding conditions (heavy car, fresh but un-optimized tires), while midfield competitors lose proportionally more performance to the same stresses.

The individual significance of Fuel_Mass_Proxy for both tiers ($p = 0.0009$ Podium, $p = 0.0003$ Point Scorer) with negative coefficients reflects the physical reality that lighter cars are faster. As fuel burns off and the proxy decreases, the log-odds of being in a higher tier increase. However, this variable is partially confounded with race progression (lap number), making the interaction term a cleaner measure of the tire–fuel coupling.

The non-significance of the tire-compound dummies (Tire_SOFT and Tire_MEDIUM for Podium) suggests that *which* compound a driver uses matters less than *how* the tire degrades over its stint. This supports the theoretical position that degradation dynamics, rather than static

compound labels, drive performance differentiation.

Relating to Literature

The findings connect to several streams in the reviewed literature.

The Box’s M test rejection of covariance homogeneity aligns with Mendoza and Cruz (2023), who similarly found that performance-tier groups exhibit unequal covariance structures, necessitating multinomial logistic regression over discriminant analysis. This pattern appears to be general across classification problems where group means and variances co-vary.

The effectiveness of the interaction term supports the feature engineering philosophy advocated by Santos (2020): domain-specific interaction terms that represent real physical relationships outperform raw variables. The Degradation $\times$ Fuel product is not an arbitrary statistical construction—it represents a real force coupling that teams actively manage during races.

The LOGO-CV validation approach follows the recommendations of Varoquaux et al. (2019) and Wong and Chen (2023), who showed that group-aware validation produces more conservative and realistic performance estimates. Our 62.36% accuracy is therefore best read as an honest, out-of-sample estimate for grouped racing data rather than an inflated score.

The tire degradation findings are consistent with Baldi et al. (2022), who showed that degradation effects are nonlinear. Our linear tire age predictor may understate late-stint effects, which could partly explain the lower recall for the Point Scorer tier—midfield drivers in the latter half of stints may exhibit degradation patterns that the linear model cannot capture.

Study Limitations

Several limitations should be acknowledged:

1. **Fuel mass approximation.** The fuel proxy assumes a uniform linear burn-off from 110 kg to 0 kg across all teams and drivers. Actual fuel loads and consumption rates are proprietary and vary by team strategy. Teams that under-fuel their cars (running lighter than the proxy suggests) would appear slower in the model than they physically are, potentially biasing tier predictions.

2. **Three-race scope.** While the three circuits cover distinct architectural types, three races represent only 13% of the 2023 calendar. Rain-affected races, sprint weekends, high-degradation circuits like Silverstone, and high-downforce tracks like Hungary are not represented. The model’s generalizability to these conditions is unknown.
3. **Unmodeled race dynamics.** DRS trains, safety car deployments, virtual safety car periods, undercut and overcut strategies, and team orders are not explicitly modeled. These factors create significant variance in the mid-field, which likely explains the low Point Scorer recall (0.45).
4. **Linear tire age.** Tire degradation is known to follow a nonlinear trajectory with a performance “cliff” after a critical wear threshold (Baldi et al., 2022). The linear tire age variable cannot capture this nonlinearity, potentially understating the effect of late-stint degradation on finishing tier.
5. **DNF protocol limitation.** Reassigning all DNF laps to “No Points” removes potentially informative telemetry from drivers who were still running competitively before retiring. A more nuanced approach, such as using running position at the time of retirement, could improve model training.

Practical Impacts

Despite its limitations, the model has practical relevance:

- **Strategy validation:** The significance of the interaction term provides quantitative evidence that tire–fuel management is a statistically distinct driver of podium outcomes. Teams can use this to validate strategic simulations that model these forces jointly rather than independently.
- **Feature engineering template:** The mean-centered interaction approach is transferable to any domain where multiple physical processes co-occur and compound—battery degradation and temperature in EVs, material fatigue and load stress in structural engineering, or player fatigue and match intensity in team sports.

- **Modeling baseline for F1 analytics:** The 62.36% LOGO-CV accuracy establishes a transparent, interpretable baseline against which more complex models (gradient-boosted trees, neural networks) can be benchmarked. Any black-box model should clear this bar while demonstrating that its additional complexity is justified.

Chapter VI

CONCLUSION

Summary of Results

This study predicted Formula One race finishing tiers (Podium, Point Scorer, No Points) using within-race telemetry data from the 2023 Bahrain, Saudi Arabian, and Australian Grands Prix. A dataset of 2,880 lap-level observations was constructed from FastF1 telemetry, with features engineered to represent the physical dynamics of race strategy: tire age, a linear fuel burn-off proxy, their mean-centered interaction term, and tire compound dummies, with grid position as a baseline control.

Box’s M test rejected the homogeneity of covariance matrices assumption ($p < 2.2 \times 10^{-16}$). This ruled out discriminant analysis and motivated multinomial logistic regression as the primary inferential model. VIF diagnostics confirmed the absence of problematic multicollinearity (all VIF < 3.5). The MLE model identified GridPosition ($p = 0.000$) and the Degradation $\times$ Fuel interaction term ($p = 0.0095$ for Podium) as statistically significant predictors. Leave-one-group-out cross-validation across the three races yielded 62.36% overall accuracy, with podium recall reaching 89%.

Connection to Literature

The findings are consistent with and extend several threads in the existing literature. The dominance of GridPosition as a predictor aligns with prior F1 prediction studies (Kuhn et al., 2022; Song & Park, 2020) that identified starting position as the primary determinant of finishing order. The rejection of covariance homogeneity mirrors findings by Mendoza and Cruz (2023) in educational performance classification, suggesting this pattern is general across multi-tier classification problems.

The central contribution—the significance of the Degradation $\times$ Fuel interaction term—extends the tire degradation literature (Baldi et al., 2022; Perdomo et al., 2021) and fuel load research (Oldknow & Gallagher, 2022) by demonstrating that the *coupling* of these two forces, not just their individual effects, carries statistically measurable predictive power. This supports

the feature engineering philosophy of Santos (2020) and provides empirical backing for the physical interaction models used in race strategy simulations.

The LOGO-CV validation approach responds directly to the methodological concerns raised by Varoquaux et al. (2019) about overly optimistic performance estimates in grouped data. By achieving 62.36% accuracy under strict cross-circuit validation, the study demonstrates that the statistical relationships are genuine and not artifacts of data leakage.

Future Research Directions

1. **Expand the race calendar.** Including all 23 races from a full F1 season would provide greater variance in weather conditions, circuit types, and team performance trajectories, strengthening both statistical power and generalizability.
2. **Incorporate weather variables.** Rain probability, track temperature, and ambient temperature directly affect tire degradation rates and compound selection. These would capture a significant source of within-race variance.
3. **Model non-linear tire degradation.** The linear tire age predictor could be replaced with a piecewise or polynomial term to capture the “cliff” effect where tire performance drops sharply after a critical wear threshold (Baldi et al., 2022).
4. **Include traffic and DRS effects.** Lap time delta to the car ahead, DRS activation history, and pit stop timing would help explain mid-field variance and improve Point Scorer tier predictions.
5. **Benchmark alternative classifiers.** Random forests, gradient-boosted trees, and ordinal logistic regression should be compared against the multinomial baseline to determine whether more flexible or order-aware models improve classification.
6. **Apply to other racing series.** The feature engineering framework—fuel proxy, tire degradation, interaction terms, grid position control—is transferable to Formula 2, IndyCar, WEC, and other series with telemetry data.

Recommendations

1. F1 analytics teams should model tire degradation and fuel mass as an interaction rather than as independent effects when building race strategy simulations and real-time prediction systems.
2. Researchers working with grouped motorsport data should adopt LOGO-CV rather than random k -fold cross-validation to produce honest out-of-sample performance estimates.
3. Feature engineering in sports analytics should prioritize physically meaningful interaction terms over raw variables, as demonstrated by the Degradation $\times$ Fuel term's significance even after controlling for baseline pace.
4. Future studies should incorporate dynamic race state variables (safety car status, DRS availability, tire compound age distribution across the field) to improve mid-field prediction accuracy.

The null hypothesis—that tire degradation, fuel mass, their interaction, and tire compound selection have no significant effect on finishing tier log-odds after controlling for grid position—is rejected. The Degradation $\times$ Fuel interaction term ($p = 0.0095 < \alpha = 0.05$) provides statistically significant evidence that the physical coupling of tire wear and fuel burn-off is a distinct, measurable predictor of podium finishes in Formula One.

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Chapter I

APPENDICES

Appendix A: Dataset Summary

The processed dataset (`f1_clean_data.csv`) contains 2,880 observations with the following variables:

Table 9: Dataset Variable Summary

Variable	Type	Values / Range
Race_ID	Categorical	2023_Bahrain, 2023_SaudiArabia, 2023_Australia
Driver	Categorical	20 unique drivers
LapNumber	Continuous	1–58
Finishing_Tier	Ordinal (3)	Podium, Point Scorer, No Points
GridPosition	Continuous	1–20
Tire_Age	Continuous	1–46
Fuel_Mass_Proxy	Continuous	5.79–108.07
Degradation_x_Fuel	Continuous	–1559 to 1559 (approx.)
Tire_MEDIUM	Binary	0 or 1
Tire_SOFT	Binary	0 or 1

Appendix B: Class Distribution

Table 10: Finishing Tier Distribution

Tier	Laps	Percentage
No Points	1,468	50.97%
Point Scorer	983	34.13%
Podium	429	14.90%
Total	2,880	100%

Appendix C: R Statistical Output

Box's M Test:

	Chi-Sq (approx.)	DF	p-value
Box's M	160.419	20	< 2.2e-16

VIF Output:

	GridPosition	Tire_Age	Fuel_Mass_Proxy
VIF	1.007456	3.219551	2.314596

	Degradation_x_Fuel	Tire_MEDIUM	Tire_SOFT
VIF	1.724701	1.181336	1.155771

Multinomial Logistic Regression Summary:

Call:

```
multinom(formula = Finishing_Tier ~ GridPosition + Tire_Age +
  Fuel_Mass_Proxy + Degradation_x_Fuel + Tire_MEDIUM +
  Tire_SOFT, data = df)
```

Coefficients:

	(Intercept)	GridPosition	Tire_Age	Fuel_Mass_Proxy
Podium	4.856516	-0.7977605	0.029083324	-0.014936022
Point Scorer	2.922957	-0.2270777	-0.003556666	-0.008448375

	Degradation_x_Fuel	Tire_MEDIUM	Tire_SOFT
Podium	0.0008256694	0.2267218	0.2424192
Point Scorer	0.0002182545	-0.4600896	-0.2538360

Residual Deviance: 3821.657

AIC: 3849.657

Exact *p*-Values:

	(Intercept)	GridPosition	Tire_Age	Fuel_Mass_Proxy
Podium	0.0000	0.0000	0.0328	0.0009
Point Scorer	0.0000	0.0000	0.6337	0.0003

	Degradation_x_Fuel	Tire_MEDIUM	Tire_SOFT
Podium	0.0095	0.3883	0.3764

Point Scorer	0.2174	0.0008	0.0540
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Appendix D: Python ML Validation Output

--- CLASSIFICATION HIT RATIO (UNSEEN RACES) ---

Overall Accuracy: 62.36%

--- CLASSIFICATION REPORT ---

	precision	recall	f1-score	support
No Points	0.74	0.67	0.70	1468
Podium	0.57	0.89	0.69	429
Point Scorer	0.49	0.45	0.47	983
accuracy			0.62	2880
macro avg	0.60	0.67	0.62	2880
weighted avg	0.63	0.62	0.62	2880